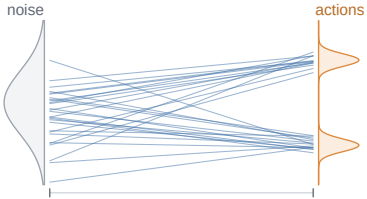
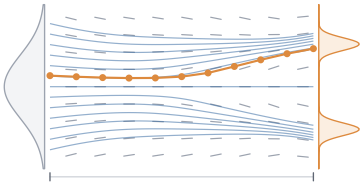
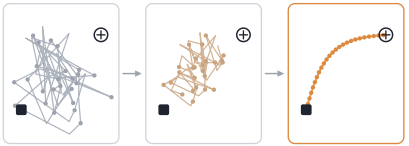
(a) Flow matching: from noise to an action chunk



training: straight paths between random (noise, action) pairs

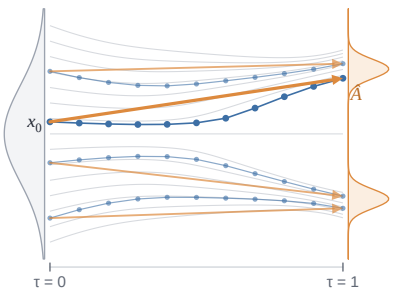


sampling: follow the learned field, N = 10 steps (dots)



2-D gripper path of one chunk (■ start, ⊕ target)

(b) Distillation: N steps → 1 step

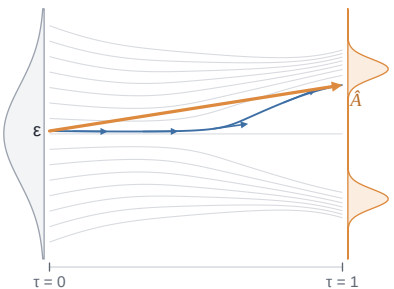


teacher: 10 Euler steps — student: 1 step

$$L(\varphi) = \|S_\varphi(x_0, o) - \Phi^{(N)}_\theta(x_0, o)\|^2$$

NFE: N → 1, needs a trained teacher

(c) One-step objective: mean velocity

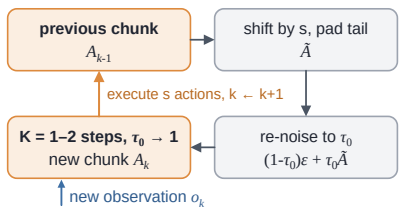


instantaneous v (tangent) — average u (chord)

one step: $\hat{A} = \varepsilon + u_\theta(\varepsilon, 0, 1 \mid o)$

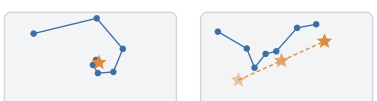
learn the average velocity directly: no teacher, NFE = 1

(d) Warm start and cross-step iteration



$$\|T_o(A) - T_o(A')\| \le \rho\|A - A'\|, \quad \rho < 1$$

holds only while the observation o is fixed



fixed o: converges to $A^*(o)$ — o_k changes: $A^*(o_k)$ moves